Ex. No: 1A Date 17/9/2026

Packet sniffing using Raw Sockets

Aim:

To implement packet sniffing using raw sockets in Python to capture and analyze network packets.

Program:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst

    # Determine the protocol
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol"
```

```
    #Print Packet details
    Print(f"Protocol: {protocol_name}")
    print(f"Source IP: {src_ip}")
    Print(f"Destination IP: {dst_ip}")
    Print("_" * 50)

Sniff(iface='Wi-Fi', prn=packet_callback,
                    filter="ip", store=0)
```

Output:

```
Protocol: TCP
Source IP: 57.155.141.117
Destination IP: 192.168.1.6

Protocol: TCP
Source IP: 192.168.1.6
Destination IP: 57.155.141.117

Protocol: TCP
Source IP: 57.155.141.117
Destination IP: 192.168.1.6

Protocol: TCP
Source IP: 57.155.141.117
Destination IP: 192.168.1.6

Protocol: UDP
Source IP: 192.168.1.6
Destination IP: 192.168.1.1
```

protocol : UDP
source IP : 192.168.1.6
destination IP : 192.168.1.1

protocol : UDP
source IP : 192.168.1.6
destination IP : 192.168.1.1

protocol : UDP
source IP : 192.168.1.1
Destination IP : 192.168.1.6

Result :

Thus, packet sniffing using raw sockets was implemented successfully, and network packets were captured and analyzed.